凸函数

$f(x) \geq f(y) + \nabla f(x)^T (y-x)$

① 凸函数的性质

二次下界：$f(x)$ 是 m-强凸、可微

$$f(y) \geq f(x) + \nabla f(x)^T (y-x) + \frac{m}{2}\|y-x\|^2$$

Defn（梯度 Lipschitz 连续）给定 f 是可微函数，若存在 $L>0$，有

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x-y\| \qquad \forall x,y \in \mathrm{dom} f.$$

则称 f 是梯度 Lipschitz 连续的.

Lemma（二次上界）给定 f 是可微函数，定义域 $\mathrm{dom} f = \mathbb{R}^n$，且 f 是 L-梯度 Lipschitz 连续的，则 f 有二次上界

$$f(y) \leq f(x) + \nabla f(x)^T (y-x) + \frac{L}{2}\|y-x\|^2$$

求解线性规划

目标：最大化利润　　求解：x_1 桌子产量

条件：木工 120 hr　　x_2 椅子产量

漆工 50 hr

subject to

$$\max\ z = 50x_1 + 30x_2$$ → 目标函数 objective function

$$\text{s.t.}\quad \left.\begin{aligned} 4x_1 + 3x_2 &\le 120 \\ 2x_1 + 1x_2 &\le 50 \\ x_1 &\ge 0 \\ x_2 &\ge 0 \end{aligned}\right\}$$ 约束 constraints.

矩阵形式

$$c = [50\ \ 30]^T \qquad x = [x_1\ \ x_2]^T$$

$$A = \begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix} \qquad b = \begin{bmatrix} 120 \\ 50 \end{bmatrix}$$

$$\max\ c^T x$$
$$\text{s.t}\quad Ax \le b \qquad x \ge 0$$

一般不用不等式

方便用单纯型法求解

$$\max\ c^T x$$
$$\text{s.t}\quad Ax = b$$
$$x \ge 0$$

要求:

① 目标函数求最大值
② $Ax=b$ 用等式方程
③ $x \ge 0$ 要求非负
④ $b \ge 0$ 要求非负

E.g.:

$$\max\ z = 50x_1 + 30x_2$$
$$\text{s.t.}\quad \begin{aligned} 4x_1 + 3x_2 &\le 120 \\ 2x_1 + 1x_2 &\le 50 \\ x_1 &\ge 0 \\ x_2 &\ge 0 \end{aligned}$$

$\Rightarrow$

$$\max\ z = 50x_1 + 30x_2 + 0\cdot x_3 + 0\cdot x_4$$
$$\text{s.t}\quad \begin{aligned} 120 &= 4x_1 + 3x_2 + x_3 \\ 50 &= 2x_1 + x_2 + x_4 \\ 0 &\le x_1 \\ 0 &\le x_2 \\ 0 &\le x_3 \\ 0 &\le x_4 \end{aligned}$$

x_3 x_4: 松弛变量

$$\Rightarrow \max\ c^T x$$
$$\text{s.t}\quad Ax = b$$
$$x \ge 0$$

$$c = \begin{bmatrix} 50 \\ 30 \\ 0 \\ 0 \end{bmatrix} \qquad A = \begin{bmatrix} 4 & 3 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{bmatrix} \qquad b = \begin{bmatrix} 120 \\ 50 \end{bmatrix}$$

E.g.:

$$\min\ -x_1 + 2x_2 - 3x_3$$
$$\text{s.t}\quad \begin{aligned} x_1 + x_2 + x_3 &\le 7 \\ x_1 - x_2 + x_3 &\ge 2 \\ -3x_1 + x_2 + 2x_3 &= 5 \\ x_1, x_2 &\ge 0 \end{aligned}$$

x_6, x_7: 松弛变量

$x_3 = x_4 - x_5$

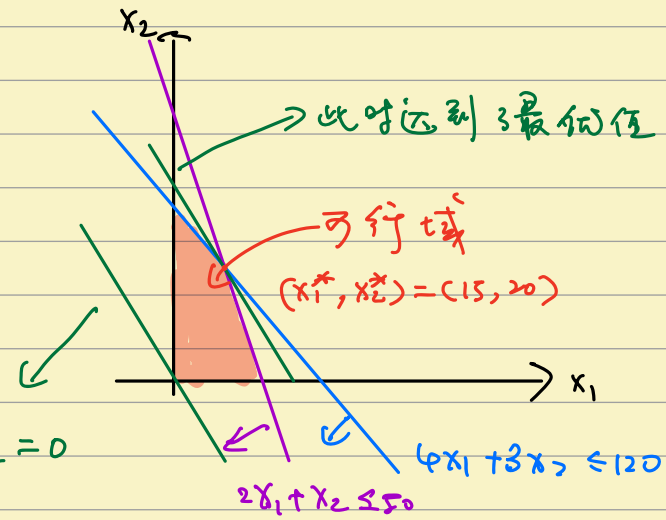
$x_3 \gtrless 0$

$\Rightarrow$ $\max \quad x_1 - 2x_2 + 3(x_4 - x_5) + 0\cdot x_6 + 0\cdot x_7$

$$
\begin{aligned}
\text{s.t.}\quad & x_1 + x_2 + x_4 - x_5 + x_6 = 7 \\
& x_1 - x_2 + x_4 - x_5 - x_7 = 2 \\
& -3x_1 + x_2 + 2x_4 - 2x_5 = 5 \\
& x_1, x_2, x_4, x_5, x_6, x_7 \geq 0
\end{aligned}
$$

经解单纯形 $\rightarrow$ **图解法**

$$
\begin{aligned}
\max \quad & 50x_1 + 30x_2 = z \\
\text{s.t.} \quad & 4x_1 + 3x_2 \leq 120 \\
& 2x_1 + x_2 \leq 50 \\
& x_1, x_2 \geq 0
\end{aligned}
$$



Note: ① 线性规划 (Linear programming) 问题 满足约束条件的可行域为多边形.

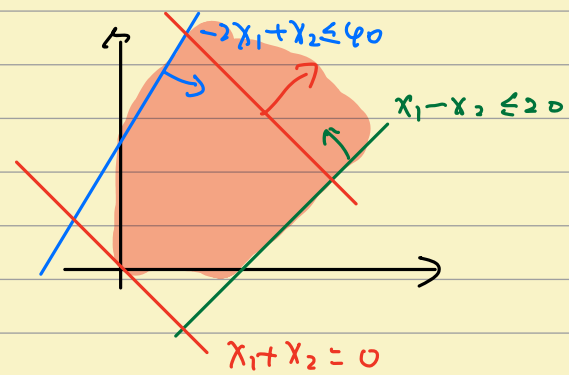
② 最优解 必定在多边形的某一个顶点取得

③ 求解结果：唯一解，无穷多最优解，无解，无界解

E.g.: $\max \quad x_1 + x_2$

$$
\begin{aligned}
\text{s.t.} \quad & -2x_1 + x_2 \leq 40 \\
& x_1 - x_2 \leq 20 \\
& x_1 \geq 0 \\
& x_2 \geq 0
\end{aligned}
$$

无界解



单纯形法 (Simplex method)

考虑:

$$\left.\begin{aligned} \max\ z &= 2x_1 + 3x_2 + 0\cdot x_3 + 0\cdot x_4 + 0\cdot x_5 \\ 8 &= x_1 + 2x_2 + x_3 \\ 16 &= 4x_1 + x_4 \\ 12 &= 4x_2 + x_5 \\ x_{1\sim5} &\geq 0 \end{aligned}\right\} \text{(LP)}$$

$$c = \begin{bmatrix} 2 \\ 3 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad A = \begin{bmatrix} 1 & 2 & 1 & 0 & 0 \\ 4 & 0 & 0 & 1 & 0 \\ 0 & 4 & 0 & 0 & 1 \end{bmatrix} \quad b = \begin{bmatrix} 8 \\ 16 \\ 12 \end{bmatrix}$$

非奇异矩阵
non-singular

$|B| \neq 0 \quad \text{rank}(B) = m$

Defn (基): A是 $m \times n$ 的矩阵, $m \leq n$, A的秩(rank)为 m
B是 A中的 m 维方阵 且非奇异矩阵, 则称 B是
LP中的一组基

$$A = \begin{bmatrix} 1 & 2 & 1 & 0 & 0 \\ 4 & 0 & 0 & 1 & 0 \\ 0 & 4 & 0 & 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 2 & 1 \\ 4 & 0 & 0 \\ 0 & 4 & 0 \end{bmatrix} \quad B' = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$B'' = \begin{bmatrix} 1 & 2 & 0 \\ 4 & 0 & 0 \\ 0 & 4 & 1 \end{bmatrix}$$

Defn (基向量): 矩阵B是由 m个线性独立的列向量组成

$$B = \begin{bmatrix} | & | & & | \\ P_1 & P_2 & \cdots & P_m \\ | & | & & | \end{bmatrix}$$

P_j 为B的基向量

$$B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad P_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad P_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad P_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Defn (基变量)：与基向量对应的变量

(非基变量)：其余的变量称为非基变量

$$A=\begin{array}{c} \begin{matrix} x_1 & x_2 & x_3 & x_4 & x_5 \end{matrix} \\ \begin{bmatrix} 1 & 2 & 1 & 0 & 0 \\ 4 & 0 & 0 & 1 & 0 \\ 0 & 4 & 0 & 0 & 1 \end{bmatrix} \end{array} \quad B=\begin{array}{c} \begin{matrix} x_3 & x_4 & x_5 \end{matrix} \\ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{array}$$

基变量：x_3 x_4 x_5

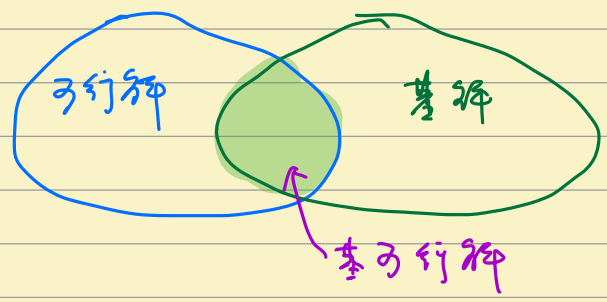
非基变量：x_1 x_2

Defn (基解)：对于某一特定的一组基 B，非基变量取0值得到的解

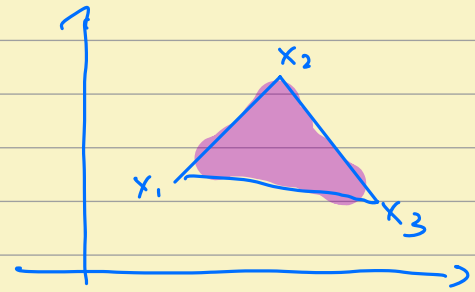
$$x=[0\ 0\ \times\ \times\ \times] \longrightarrow \text{基解}.$$

非基变量为零

Defn (基可行解) 满足非负约束条件的基解



Lemma (凸组合) 若 $\Omega=\{x \mid Ax=b,\ x\geq 0\}$，则任一点 $x\in\Omega$，可表示为顶点的凸组合



Thm：线性规划问题的基可行解对应于可行域的顶点.

Thm：若可行集有界，LP问题的目标函数在可行集顶点上达到最优

单纯形法例子.

$$\max \quad z = 2x_1 + 3x_2 + 0\cdot x_3 + 0\cdot x_4 + 0\cdot x_5$$

$$\begin{aligned} 8 &= x_1 + 2x_2 + x_3 \\ 16 &= 4x_1 + x_4 \\ 12 &= 4x_2 + x_5 \end{aligned}$$

$$x_{1\sim5} \geq 0$$

$$c = \begin{array}{c} \begin{array}{ccccc} x_1 & x_2 & x_3 & x_4 & x_5 \end{array} \\ \begin{bmatrix} 2 & 3 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

$$A = \begin{bmatrix} 1 & 2 & 1 & 0 & 0 \\ 4 & 0 & 0 & 1 & 0 \\ 0 & 4 & 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 8 \\ 16 \\ 12 \end{bmatrix} = b$$

$$B = \begin{array}{c} \begin{array}{ccc} x_3 & x_4 & x_5 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{array}$$

$$\begin{aligned} x_3 &= 8 - x_1 - 2x_2 \\ x_4 &= 16 - 4x_1 \\ x_5 &= 12 - 4x_2 \end{aligned}$$

基可行解: $x = (0 \ 0 \ 8 \ 16 \ 12)$

固定 $x_1 = 0$

$$\begin{aligned} x_3 &= 8 - x_1 - 2x_2 = 8 - 2x_2 \geq 0 \\ x_4 &= 16 - 4x_1 = 16 \geq 0 \\ x_5 &= 12 - 4x_2 = 12 - 4x_2 \geq 0 \end{aligned}$$

令 $x_2 = \min\left(\frac{8}{2}, \frac{12}{4}\right) = 3$, 则 $x_5 = 0$

基可行解: $x = (0 \ 3 \ 2 \ 16 \ 0)$

固定 $x_5 = 0$

$$\begin{aligned} x_3 + 2x_2 &= 2 - x_1 \\ x_4 &= 16 - 4x_1 \\ 4x_2 &= 12 - x_5 \end{aligned}$$

$$\Rightarrow \begin{cases} x_3 = 2 - x_1 + \frac{1}{2}x_5 = 2 - x_1 \geq 0 \\ x_4 = 16 - 4x_1 = 16 - 4x_1 \geq 0 \\ x_2 = 3 - \frac{1}{4}x_5 = 3 \geq 0 \end{cases}$$

令 $x_1 = \min\left(\frac{2}{1}, \frac{16}{4}\right) = 2$ 则 $x_3 = 0$

基可行解 $x = (2\ 3\ 0\ 8\ 0)$

$$\Rightarrow \begin{aligned} x_1 &= 2 - x_3 + \frac{1}{2}x_5 \\ x_4 &= 8 + 4x_3 - 2x_5 \\ x_2 &= 3 - \frac{1}{4}x_5 \end{aligned} \qquad x_3\ x_5$$

固定 $x_3 = 0$

$$\begin{aligned} x_1 &= 2 + \frac{1}{2}x_5 \\ x_4 &= 8 - 2x_5 \\ x_2 &= 3 - \frac{1}{4}x_5 \end{aligned}$$

$x_5 = \min\left(\frac{8}{2}, \frac{3}{\frac{1}{4}}\right) = 4 \Rightarrow x_4 = 0$

$x = (4, 2, 0\ 0\ 4)$